

Chapter 8

Lecture 8: Subspace topology. Closed sets. Interior and Closure of a set.

8.1 Subspace Topology

Definition 8.1.1. Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$. The **subspace topology** on Y is defined as

$$\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}.$$

Property 8.1.1. $(Y, \mathcal{T}_Y)$ is a topological space.

Lemma 8.1.1. If $\mathcal{B}$ is a basis for the topology $\mathcal{T}$ on X , then the collection

$$\mathcal{B}_Y = \{B \cap Y : B \in \mathcal{B}\}$$

is a basis for the subspace topology $\mathcal{T}_Y$ on Y .

Proof

Let $U \in \mathcal{T}$ and $y \in U \cap Y$. Then there exists $B \in \mathcal{B}$ such that $y \in B \subseteq U$. Then $y \in B \cap Y \subseteq U \cap Y$.

Remark The set which is open in the subspace topology may not be open in the original topology.

Property 8.1.2. If Y is open in $(X, \mathcal{T})$ and $U \subseteq Y$ is open in $(Y, \mathcal{T}_Y)$, then U is open in $(X, \mathcal{T})$.

Proof Since $U \subseteq Y$ is open in $(Y, \mathcal{T}_Y)$, there exists $V \in \mathcal{T}$ such that $U = V \cap Y$. Since Y is open in $(X, \mathcal{T})$, we have $U = V \cap Y$ is open in $(X, \mathcal{T})$.

Theorem 8.1.1. Let $A \subseteq X$ and $B \subseteq Y$ such that $A \times B \subseteq X \times Y$. Then the product of subspace topologies $\mathcal{T}_A \times \mathcal{T}_B$ is equal to the subspace topology $\mathcal{T}_{A \times B}$ on $A \times B$.

Proof

Let $U \subseteq X$ and $V \subseteq Y$ be open sets in $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ respectively. Then the products of the form $U \times V$ form a basis for the product topology on $X \times Y$. And we have

$$(U \times V) \cap (A \times B) \tag{8.1}$$

form a basis for the subspace topology on $A \times B$. Note that

$$(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B) \tag{8.2}$$

So the basis for the product of subspace topologies is equal to the basis for the subspace topology on $A \times B$. ■

Let $(X, <)$ be an ordered set. Let $a_0 = \min(X)$ and $b_0 = \max(X)$ if they exist. Then

$$\mathcal{B} = \{(a, b) : a_0 < a < b < b_0\} \cup \{[a_0, b) : b < b_0\} \cup \{(a, b_0] : a > a_0\} \tag{8.3}$$

is a basis for the order topology on X . Notice that $[a_0, b)$ refers to the set $\{x \in X : x < b\}$ when a_0 does not exist.

Remark Let $Y \subseteq X$ be a subset inheriting the order from X . Then it may happen that the order topology on Y is different from the subspace topology on Y .

Example 8.1.1. Let $X = \mathbb{R}$ with the usual order and $Y = [0, 1) \cup \{2\}$ be a subset of X . Then $\{2\}$ is open in the subspace topology on Y since $\{2\} = (1.5, 2.5) \cap Y$. However, $\{2\}$ is not open in the order topology on Y .

Exercise Show that for the order topology $[0, 1) \cup \{2\}$ is connected.

Example 8.1.2. Let $X = Y = \mathbb{R}$ with the usual order. Then the product of order topologies is the standard topology on $\mathbb{R}^2$. We take the lexicographic order as the order on $\mathbb{R}^2$.

Remark Let $I \times I = [0, 1] \times [0, 1] \subseteq X \times Y$. Then the order topology on $[0, 1] \times [0, 1]$ is different from the subspace topology inherited from the order topology on $\mathbb{R}^2$. The set $\{1/2\} \times (1/2, 1]$ is not open in the order topology on $I \times I$ but it is open in the subspace topology inherited from $\mathbb{R}^2$.

Definition 8.1.2. Y is convex if for any $a, b \in Y$, the interval $(a, b) \subseteq Y$.

Theorem 8.1.2. Let $Y \subseteq X$ be a convex subset of $(X, <)$. Then the restriction of the order topology on X to Y is equal to the order topology on Y .

Proof

Take $(a, +\infty), (-\infty, b) \subseteq X$ which form a basis for the order topology on X . Take $Y \subseteq X$ convex.

1. If $a \in Y$, then $(a, +\infty) \cap Y = \{y | y \in Y \text{ and } a < y\}$.
2. If $a \notin Y$, then there are only two cases because Y is convex:
 - (a) If a is a lower bound for Y , then $(a, +\infty) \cap Y = Y$
 - (b) If a is an upper bound for Y , then $(a, +\infty) \cap Y = \emptyset$

Then we obtain the basis for subspace topology on Y from the basis of order topology on X . And also this is the basis for the order topology on Y .

Definition 8.1.3. $A \subseteq X$ is closed if and only if $X \setminus A$ is open.

Property 8.1.3. 1. X and $\emptyset$ are closed.

2. The intersection of any collection of closed sets is closed.

3. The union of finitely many closed sets is closed.

Theorem 8.1.3. Let X be a topological space. And $Y \subseteq X$ has a subspace topology. Then $A \subseteq Y$ is closed in Y if and only if there exists a closed set C in X such that $A = C \cap Y$.

Proof

$(\Rightarrow)$ Since A is closed in Y , then $Y \setminus A$ is open in Y . So there exists an open set U in X such that $Y \setminus A = U \cap Y$. Let $C = X \setminus U$. Then C is closed in X and

$$A = Y \setminus (Y \setminus A) = Y \setminus (U \cap Y) = Y \cap (X \setminus U) = Y \cap C.$$

$(\Leftarrow)$ Let $A = C \cap Y$ where C is closed in X . Then $X \setminus C$ is open in X . So we have $(X \setminus C) \cap Y$ is open in Y . But we have

$$(X \setminus C) \cap Y = Y \setminus (C \cap Y) = Y \setminus A. \quad (8.4)$$

which is open in Y . So A is closed in Y .

Definition 8.1.4. If $U \subseteq X$ is open and $x \in U$ then U is a neighborhood of x .

Definition 8.1.5. $\text{Int}(A) = \bigcup \{U \subseteq X | U \text{ is open and } U \subseteq A\}$ is the interior of A .

$\bar{A} = \bigcap \{C \subseteq X | C \text{ is closed and } A \subseteq C\}$ is the closure of A .

Remark $\text{Int}(A) \subseteq A \subseteq \bar{A}$.

Remark If $A \subseteq Y \subseteq X$, then the closure of A in Y and the closure of A in X may be different. For example let $X = \mathbb{R}$ and $Y = [0, 1)$. Let $A = (0, 1) \subseteq Y$. Then the closure of A in Y is $[0, 1)$ while the closure of A in X is $[0, 1]$.

Theorem 8.1.4. Let $Y \subseteq X$ with the subspace topology. Let $\bar{A}$ be the closure of A in X . Then the closure of A in Y is equal to $\bar{A} \cap Y$.

Proof

Let B be the closure of A in Y . We prove the two inclusions.

Let B be the closure of A in Y . $\bar{A}$ is closed in X . So $\bar{A} \cap Y$ is closed in Y . Then $\bar{A} \cap Y \supseteq A$ (Since we are talking about the closure in Y , we must have $A \subseteq Y$). So $B \subseteq \bar{A} \cap Y$ (B is the smallest closed set containing A).

Since B is closed in Y , there exists a closed set C in X such that $B = C \cap Y$. Hence $C \supseteq A$ which is closed in X . So $C \supseteq \bar{A}$. Then $\bar{A} \cap Y \subseteq C \cap Y = B$.

Hence the two inclusions imply that $B = \bar{A} \cap Y$. ■